ANAT 1009 – Lab #3: Muscular System

Purpose:

- To explore the microscopic structure of muscle tissue types
- To develop an understanding of the location and function of various skeletal muscles throughout the human body

Video Available: Muscular System Introduction

The Appendix, located at the end of the lab, lists the structures for which you are responsible.

Station 1: Types of Muscle Tissue

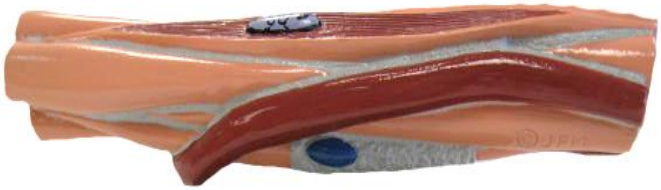
1. Identify the three different types of muscle tissue.



a



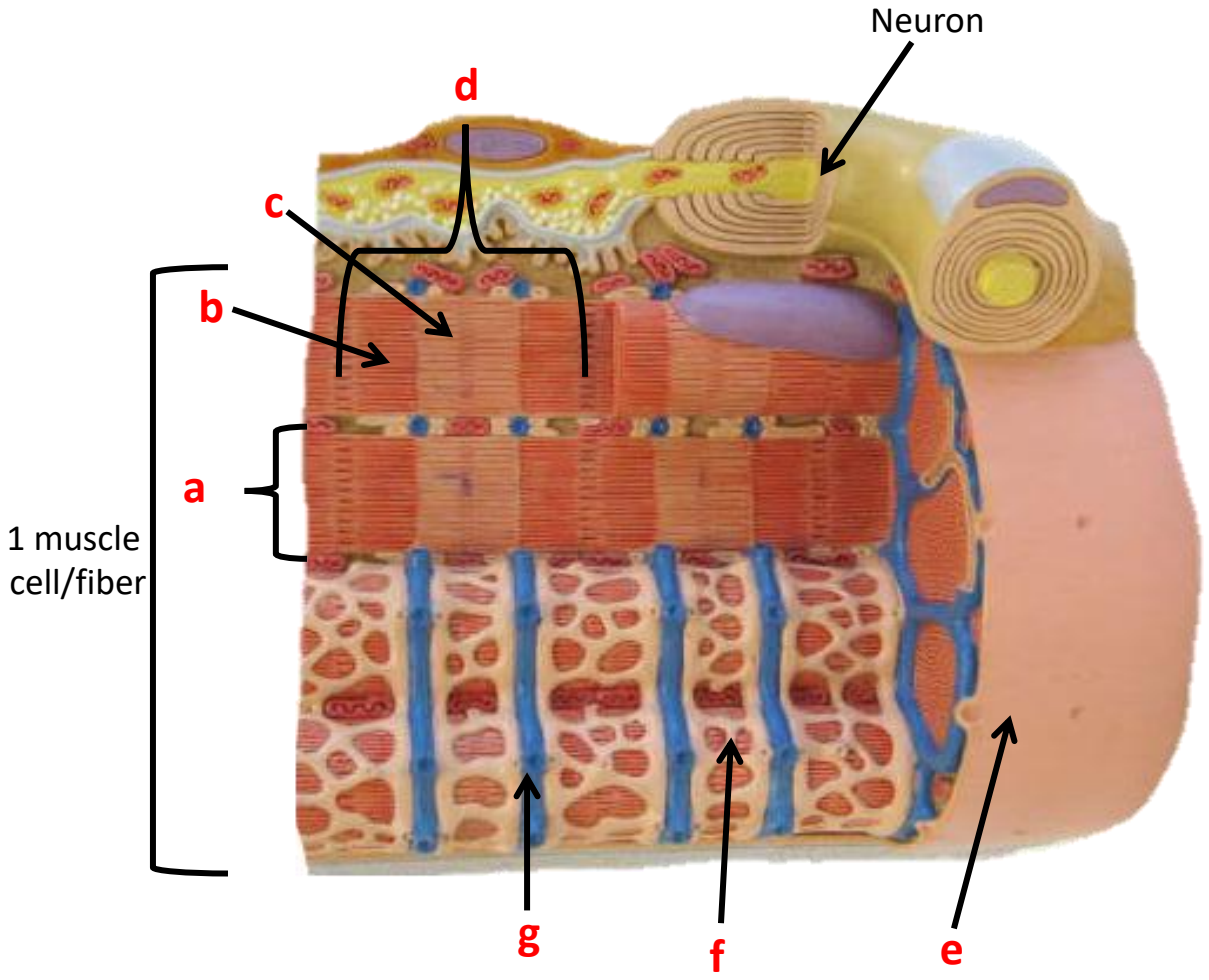
b



c

Station 2: Structure of skeletal Muscle

1. Identify structures listed in the appendix on the model below.

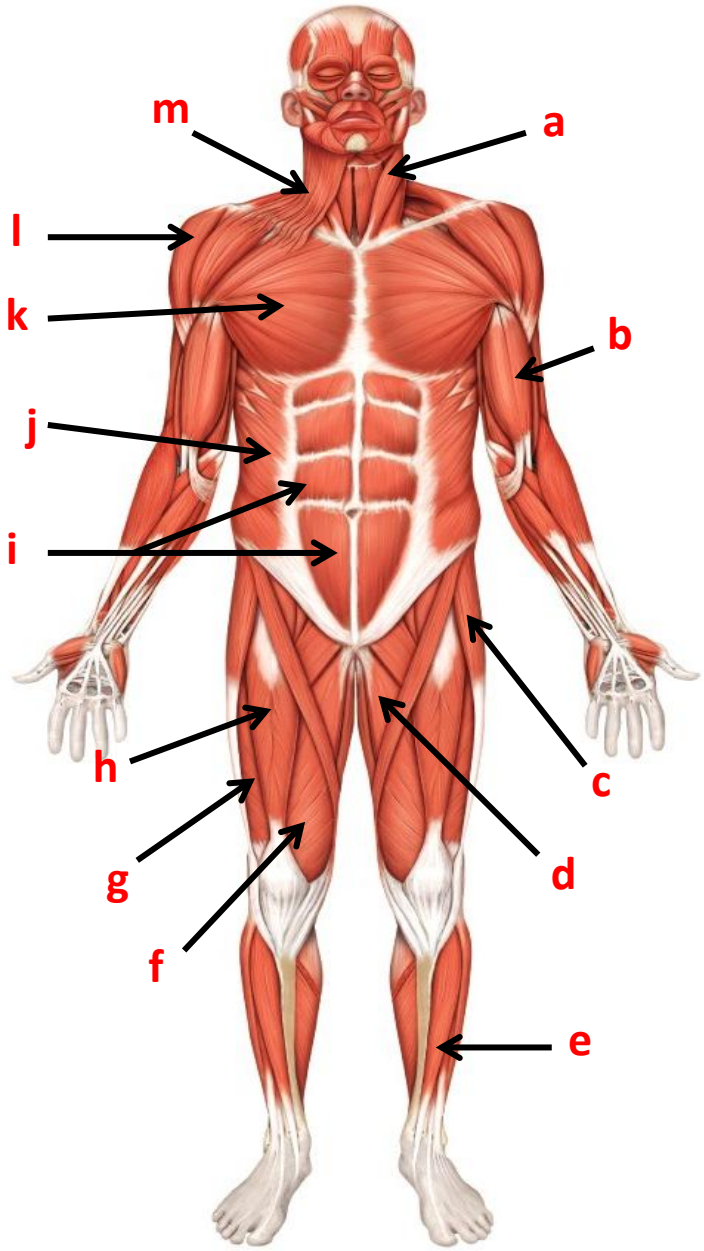


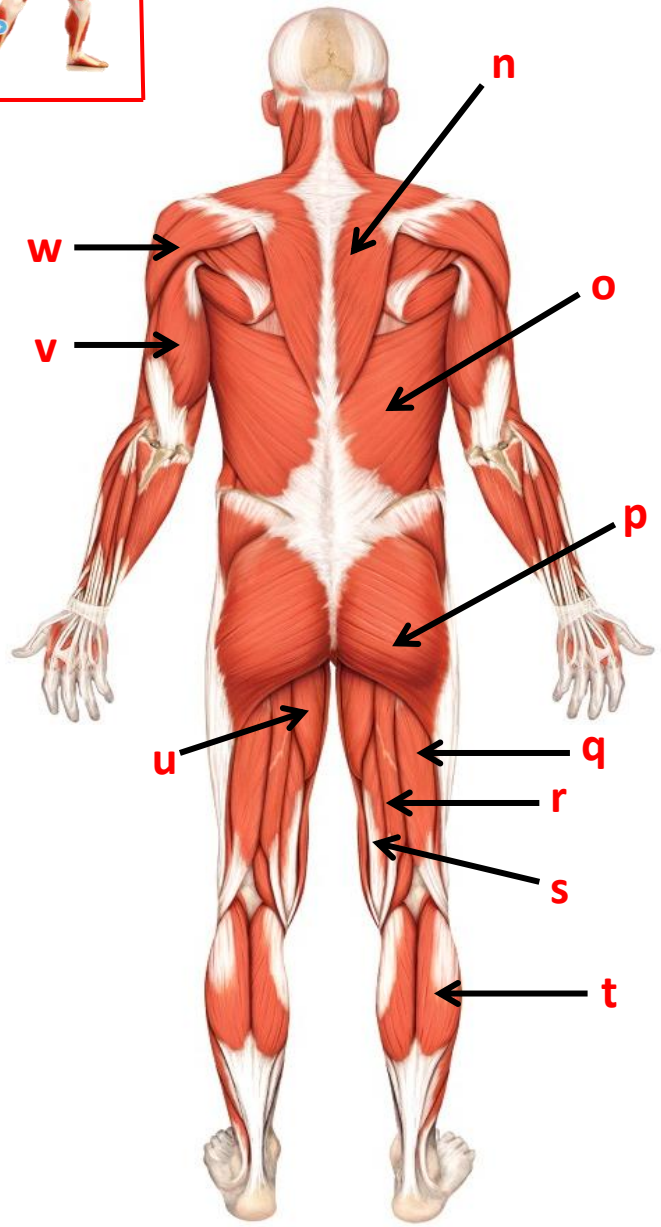
Station 3: General Musculature

1. Identify and label the superficial muscles listed in the appendix on the figure below. NOTE: not all of the muscles you are responsible are shown on this station; be sure to complete the other muscle stations.



Rotational Image Available: Rotational Images -> Muscular System -> Whole Body Musculature



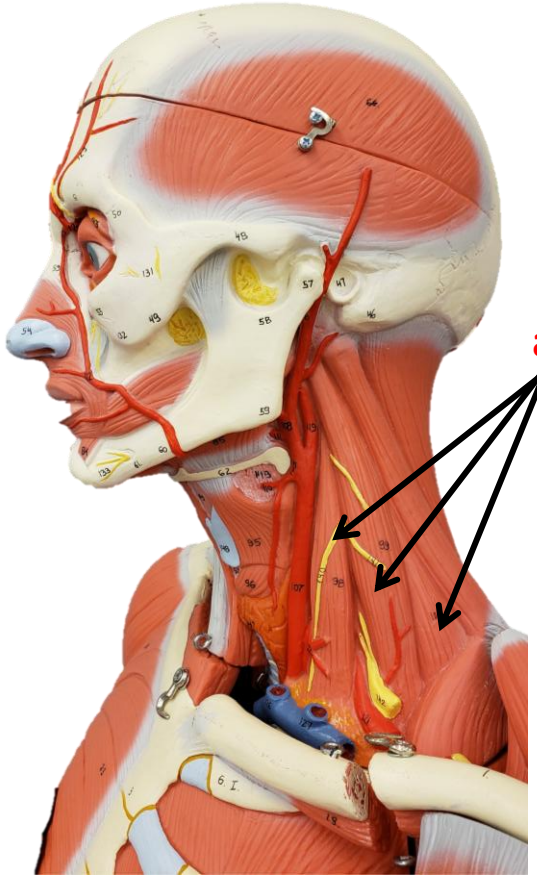


Station 4: Head and Neck Musculature

1. Identify and label the muscles listed in the appendix on the appropriate images below.



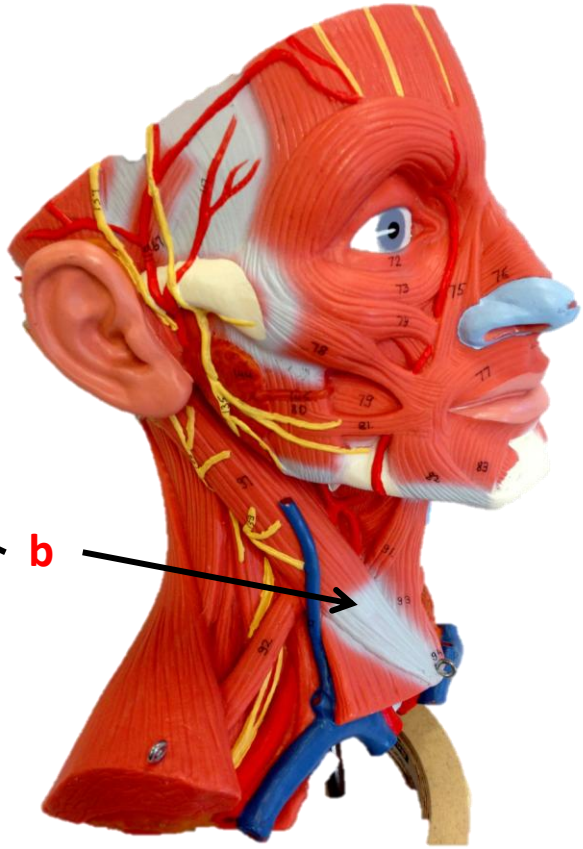
Rotational Images Available: Rotational Images -> Muscular System -> Sagittal Head and Neck & Head, and Neck Musculature



a



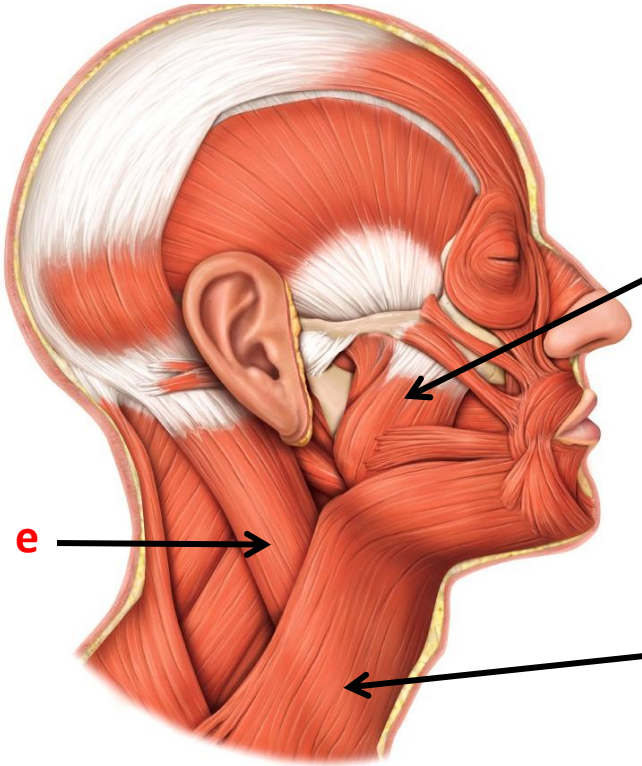
b

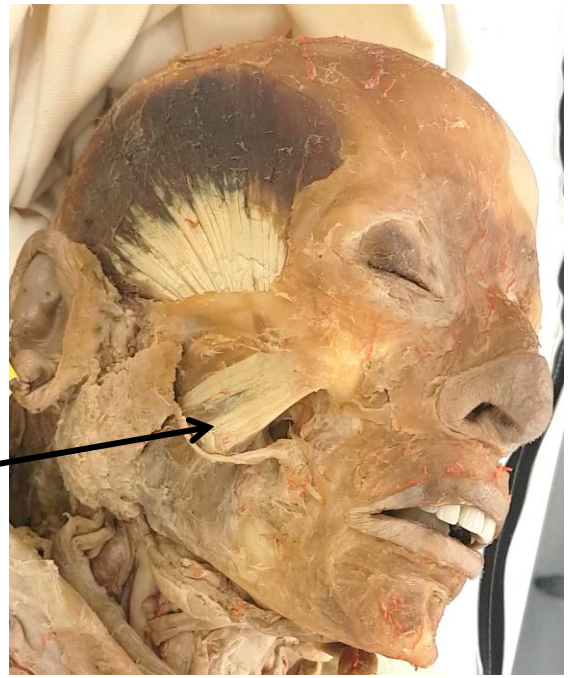


Station 4: Head and Neck Musculature (continued)

1. Identify and label the muscles listed in the appendix on the appropriate images below.

Videos Available:
Muscles of the neck
Muscles of the face



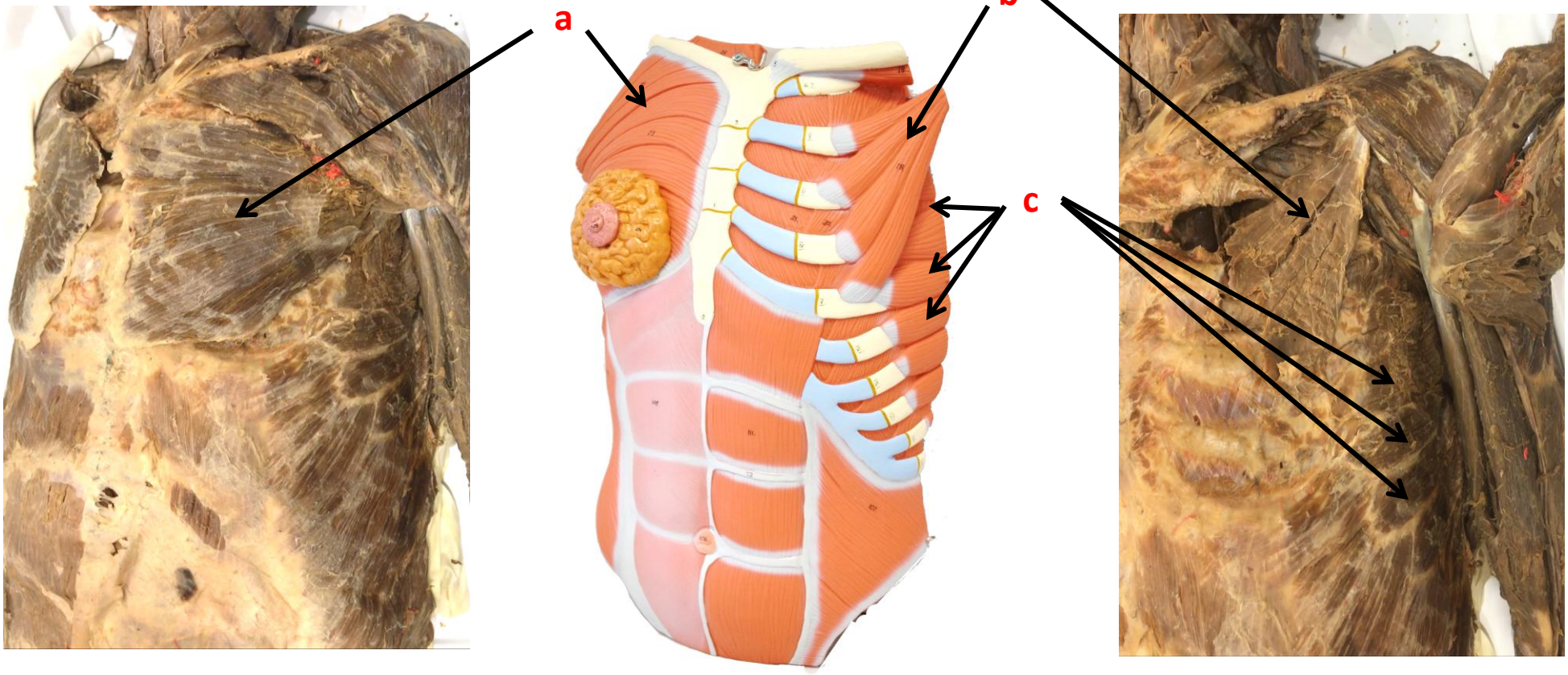




Station 5: Anterior Thoracic and Abdominal Musculature

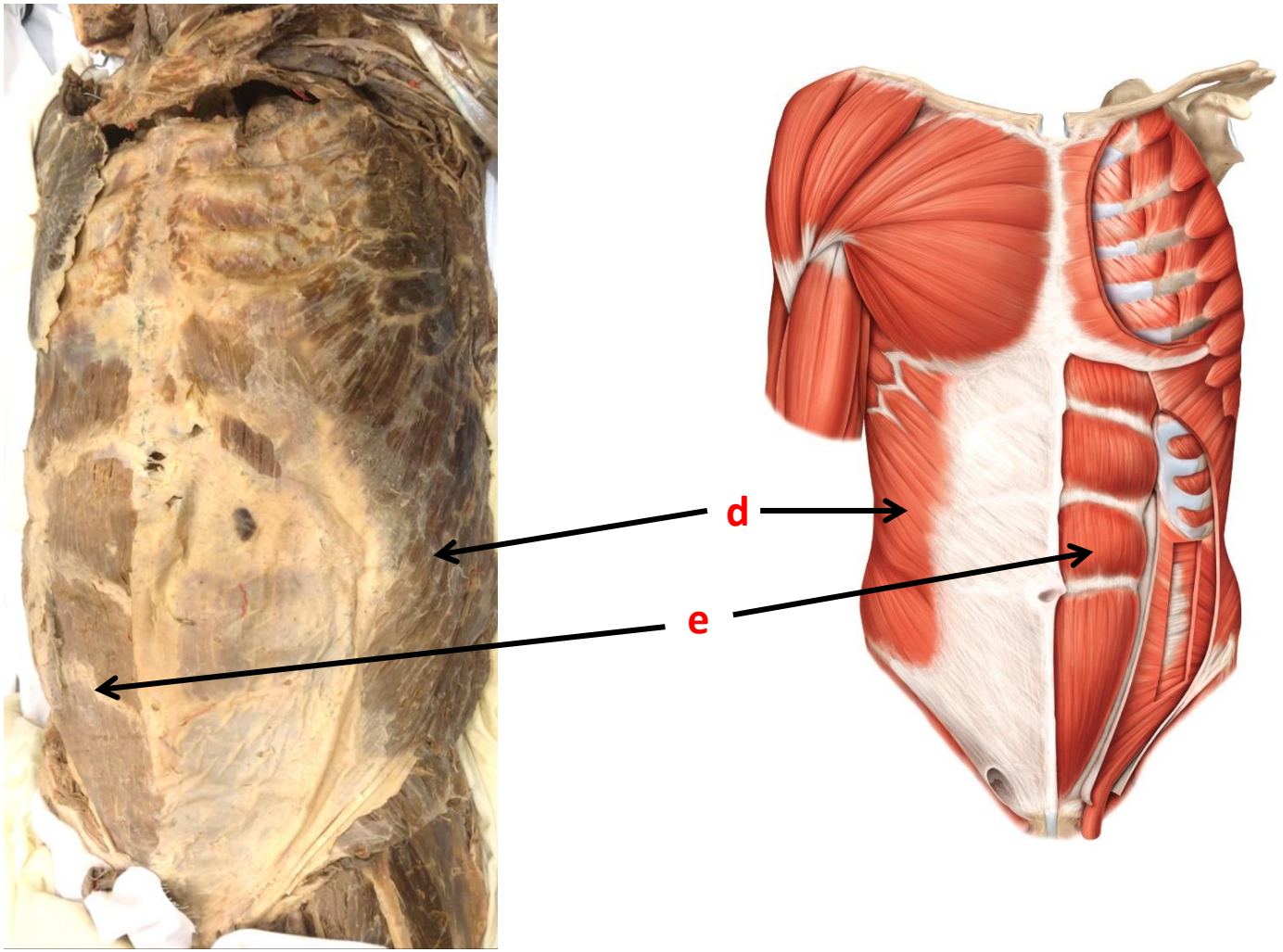
1. Identify and label the muscles listed in the appendix on the appropriate images below.

Video Available:
Muscles of the chest wall and abdomen



Station 5: Anterior Thoracic and Abdominal Musculature (continued)

1. Identify and label the structures listed in the appendix on the appropriate images below.

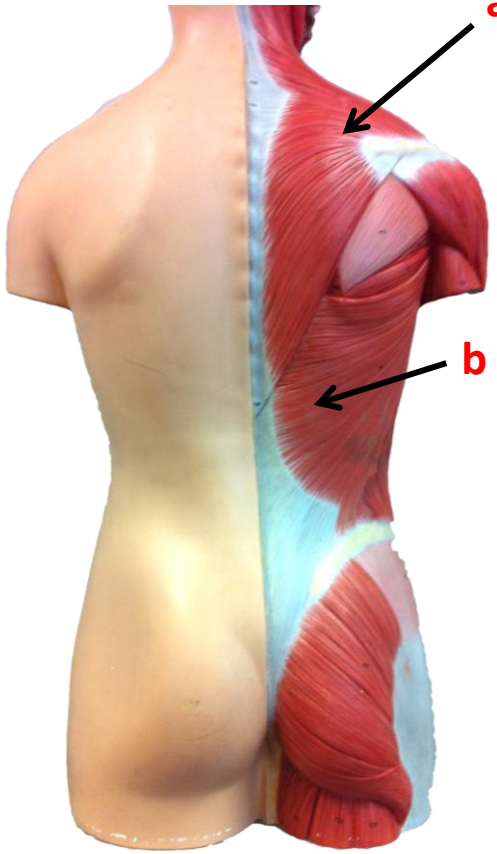


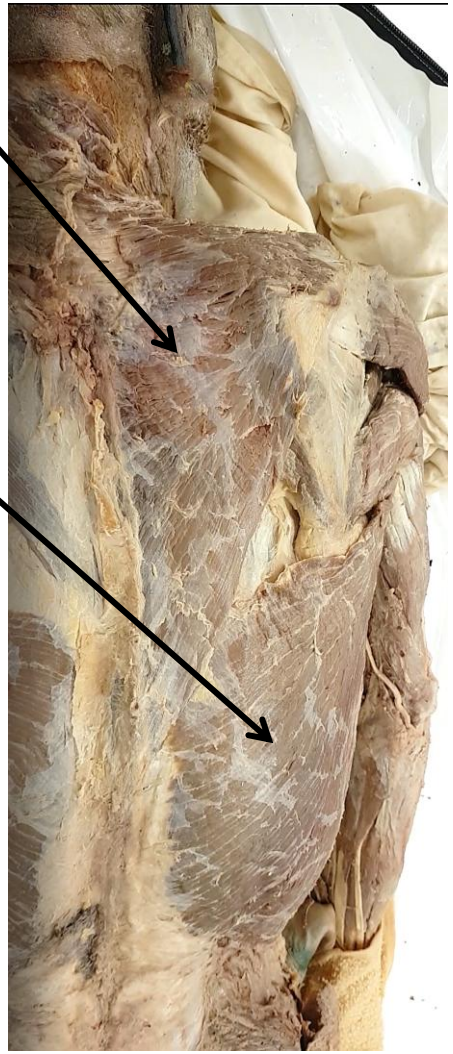
Copyright © John Wiley & Son Inc.

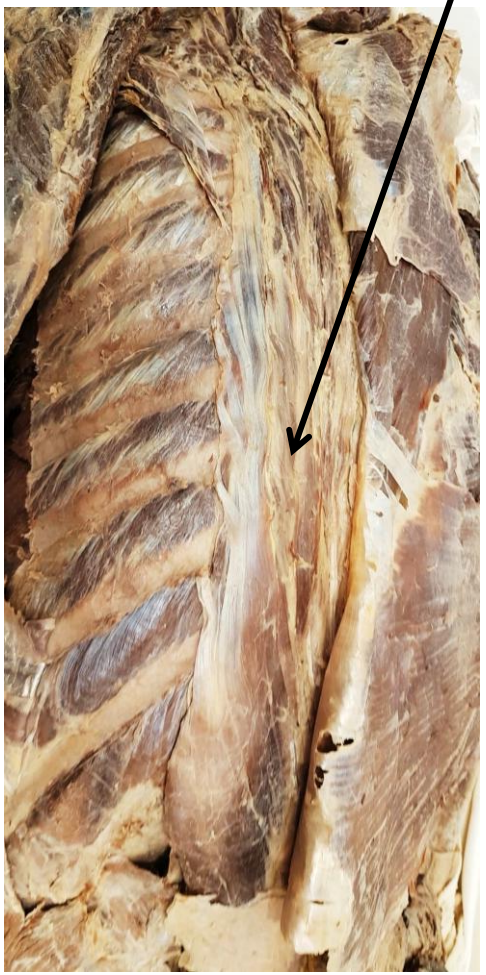
Station 6: Upper and Lower Back Musculature

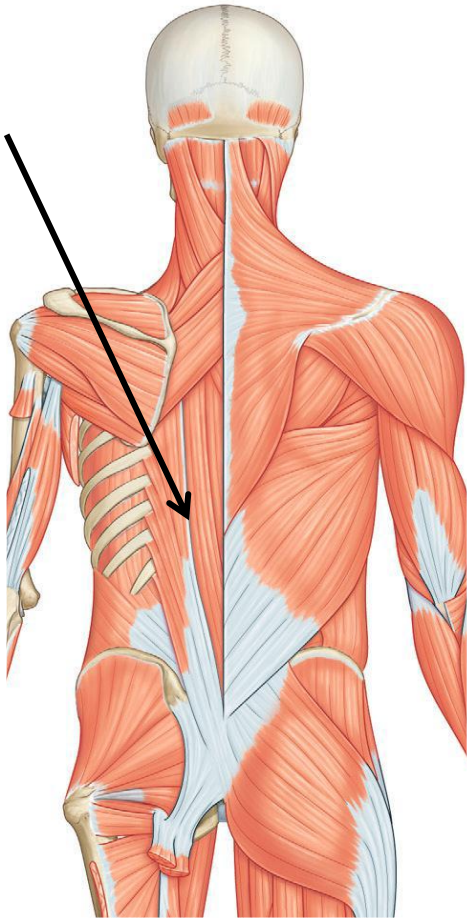
1. Identify and label the muscles listed in the appendix on the appropriate images below.

Video Available:
Muscles of the back



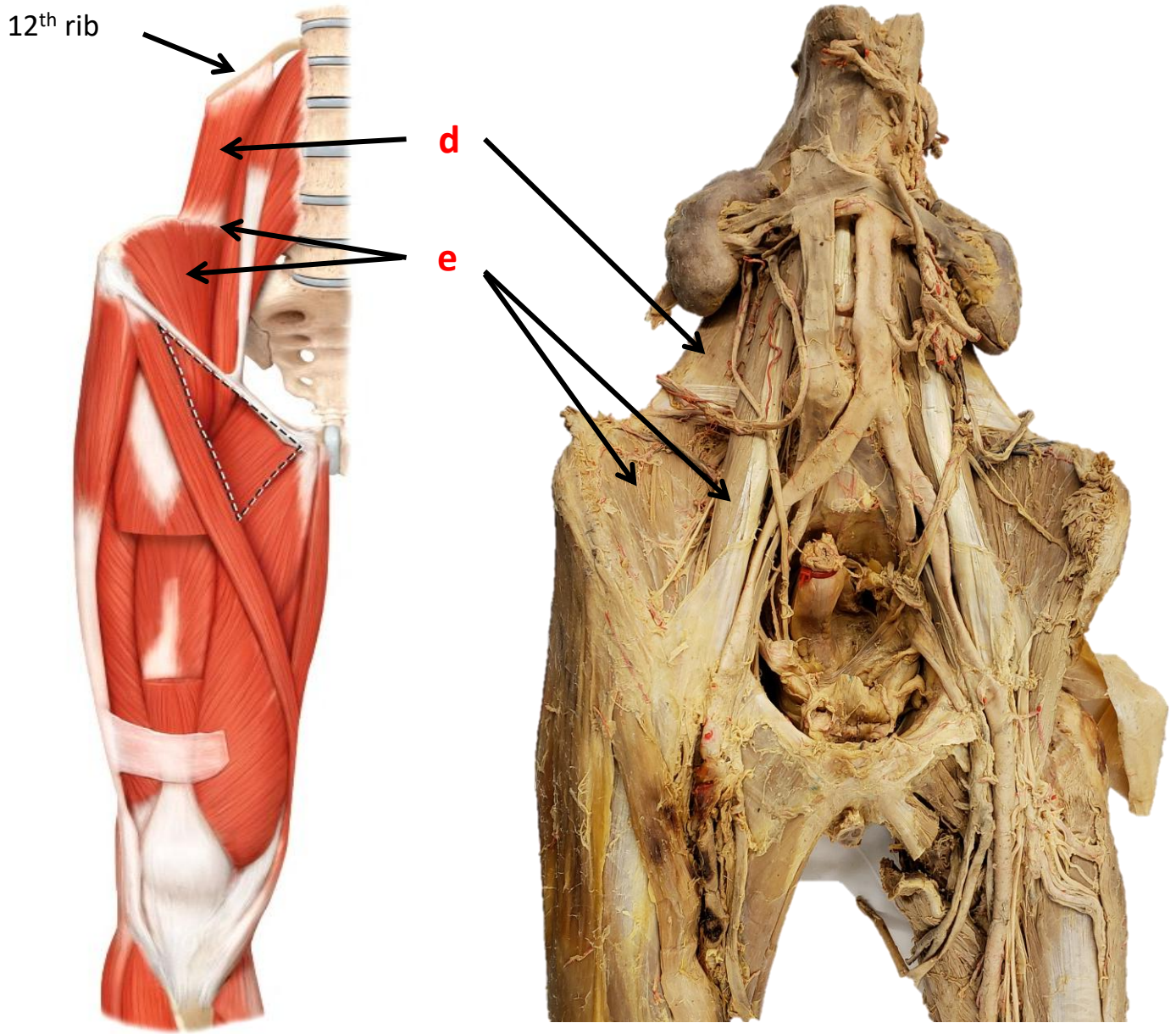






Station 6: Upper and Lower Back Musculature (continued)

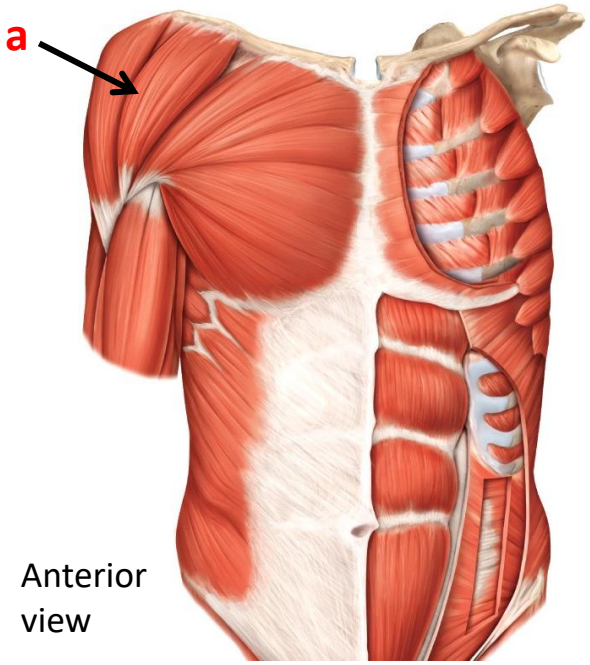
1. Identify and label the muscles listed in the appendix on the appropriate images below.



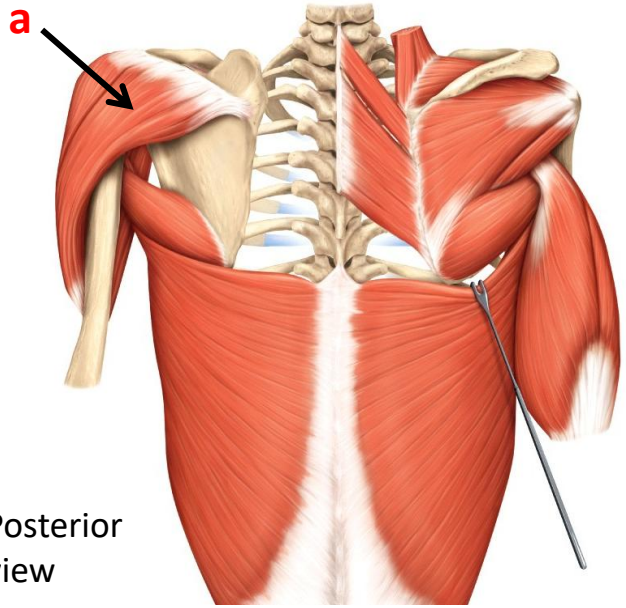
Station 7: Arm Musculature

1. Identify and label the structures listed in the appendix on the appropriate images below.

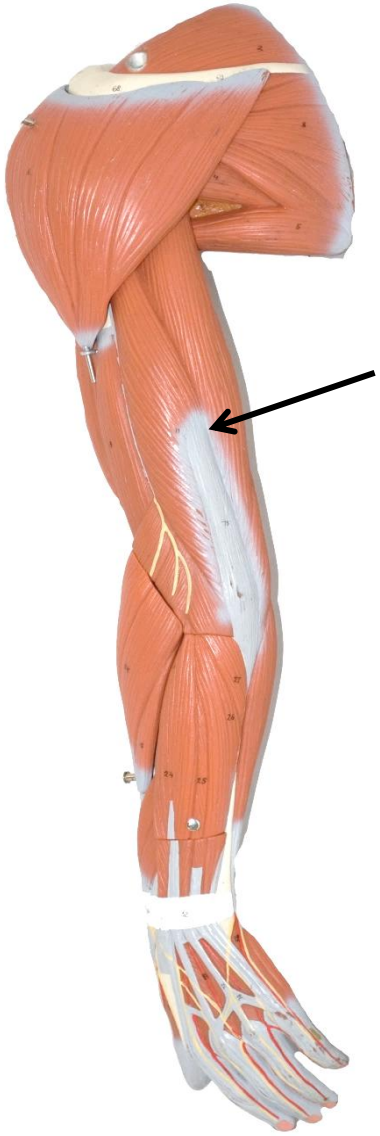
Video Available:
Posterior arm muscles



Anterior view



Posterior view



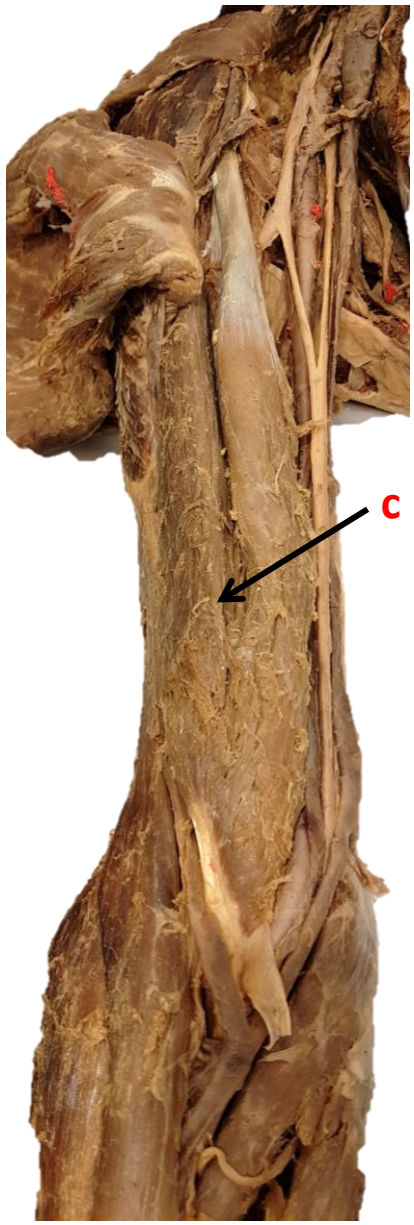


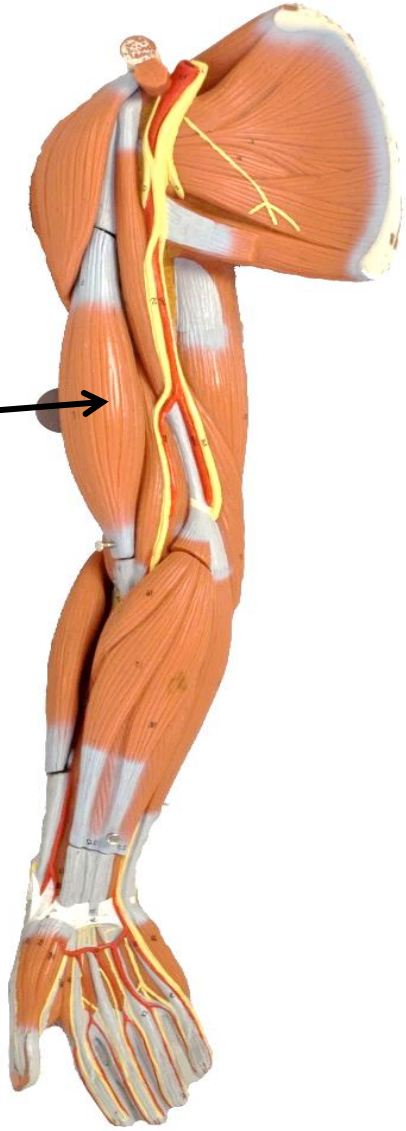
Posterior view

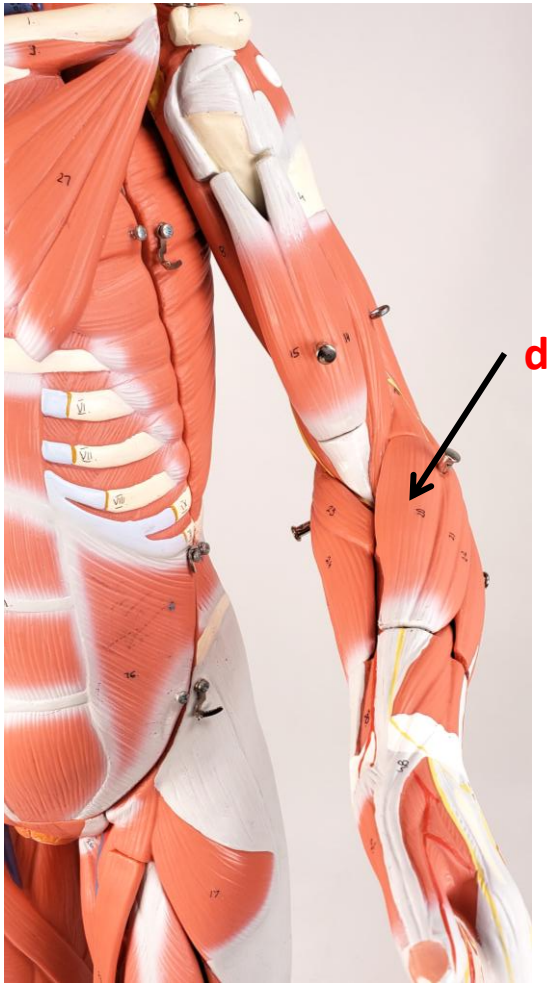
Station 7: Arm Musculature (continued)

1. Identify and label the structures listed in the appendix on the appropriate images below.

Video Available:
Anterior arm muscles





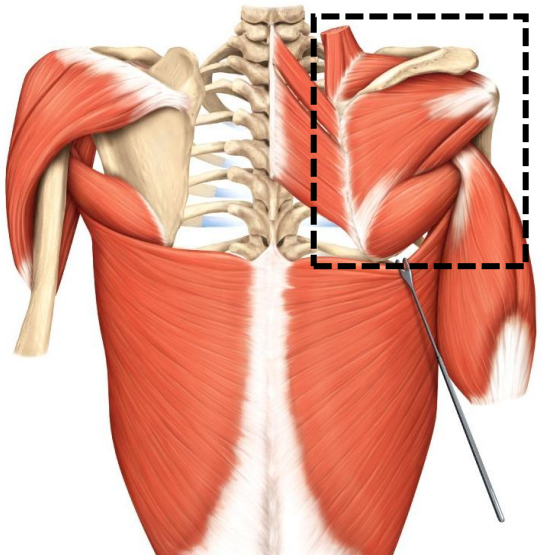




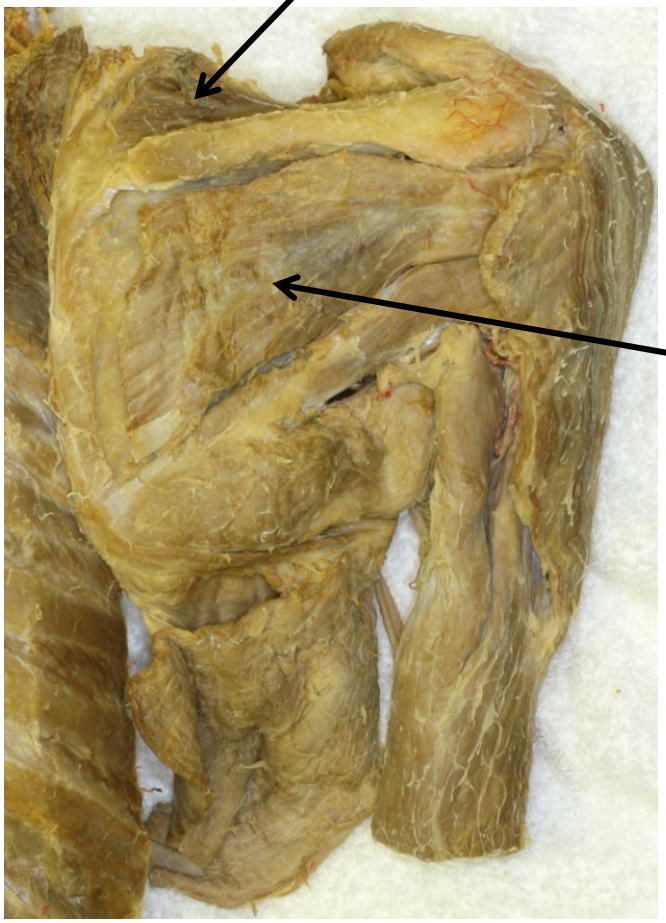
Anterior view

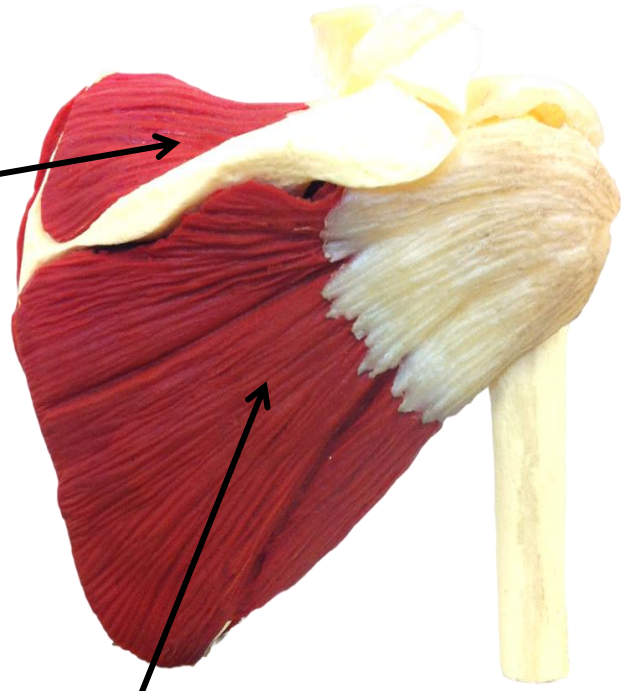
Station 7: Arm Musculature (continued)

1. Identify and label the structures listed in the appendix on the appropriate images below.



Posterior view

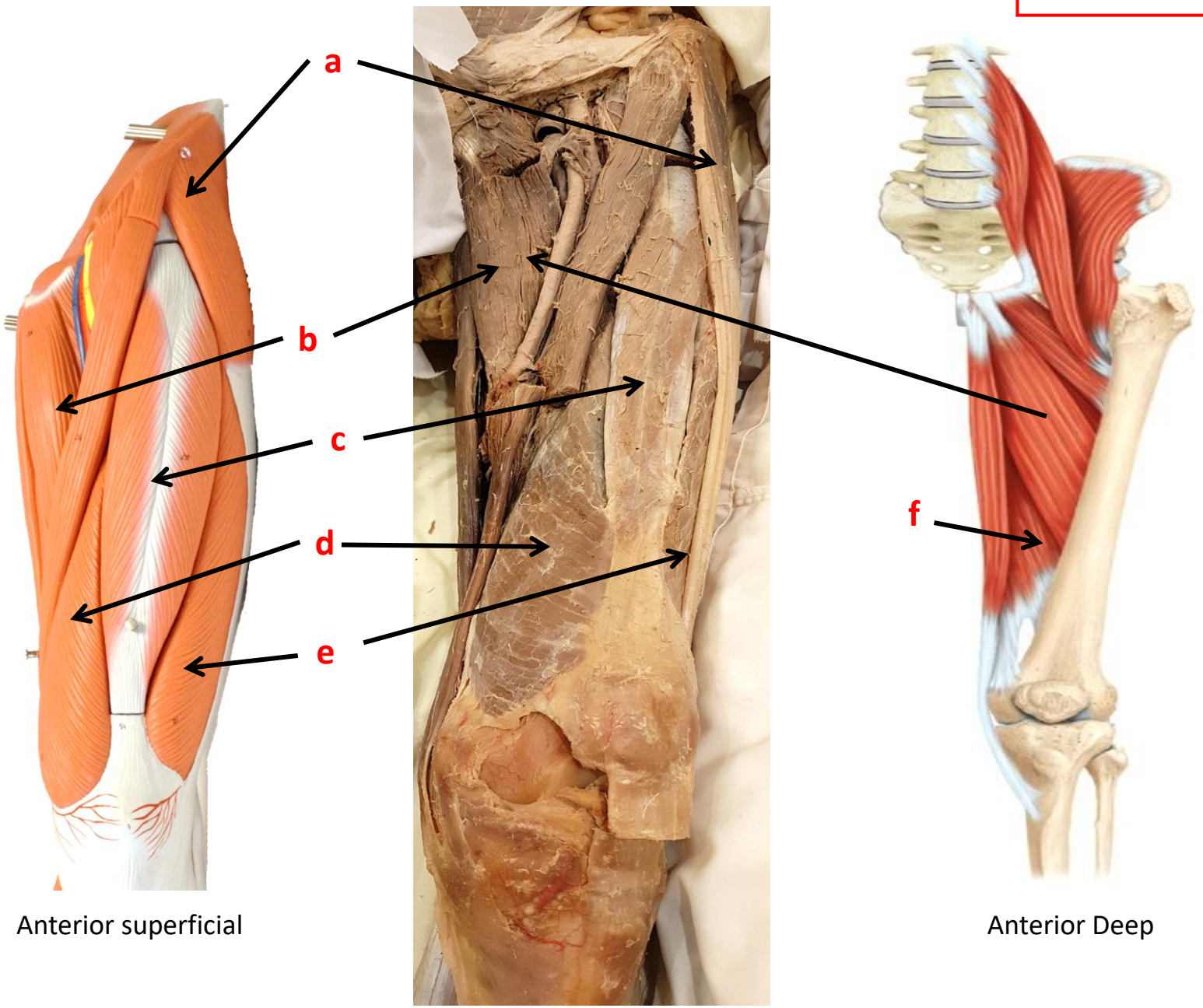




Station 8: Musculature of the Lower Limb

1. Identify and label the structures listed in the appendix on the diagrams below.

Video Available:
Muscles of the anterior and medial thigh



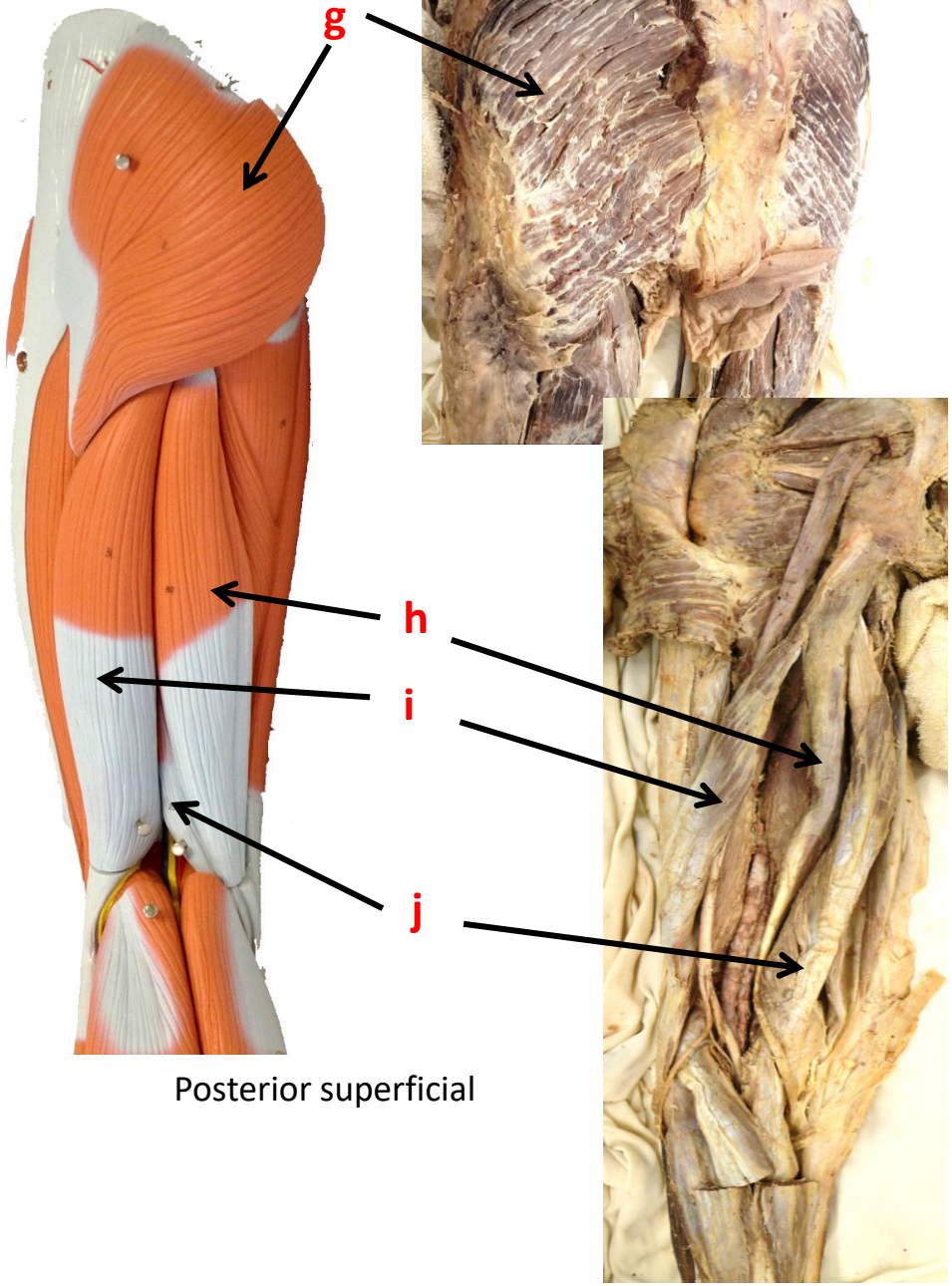
Anterior superficial

Anterior Deep

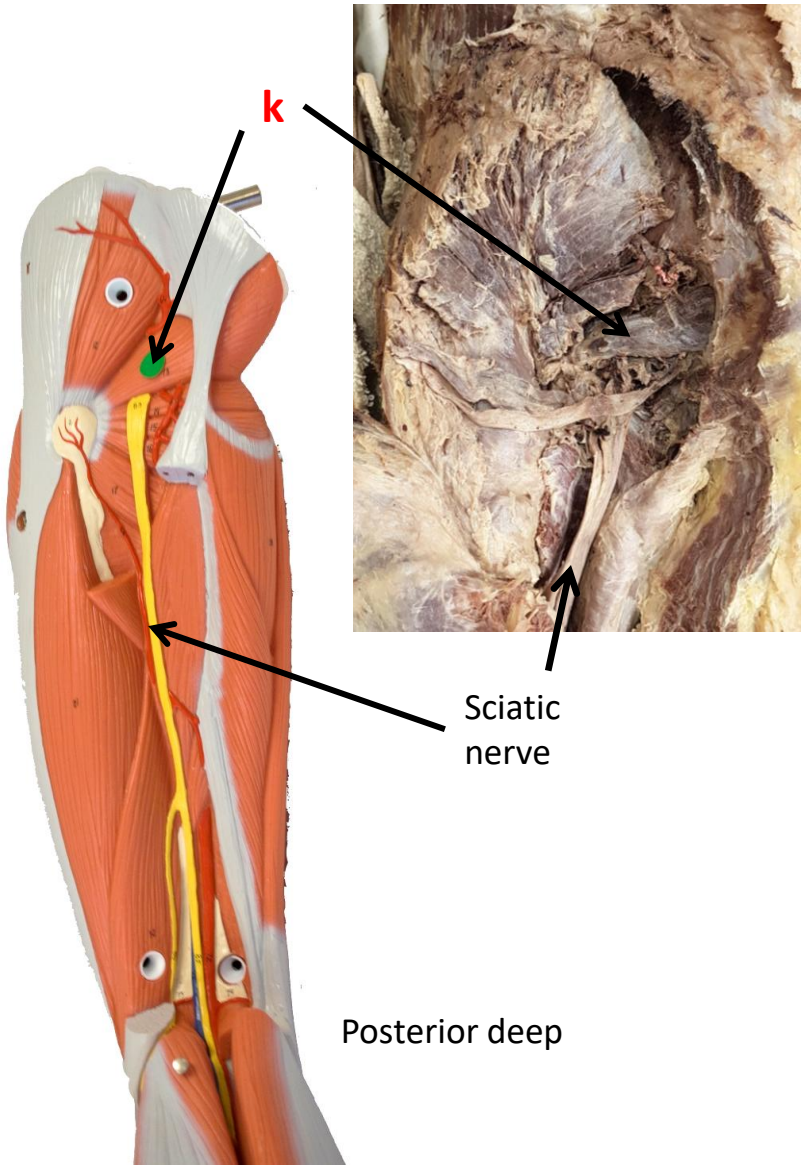
Station 8: Musculature of the Lower Limb (continued)

1. Identify and label the structures listed in the appendix on the diagrams below.

Video Available:
Muscles of the gluteal region, posterior thigh & posterior leg



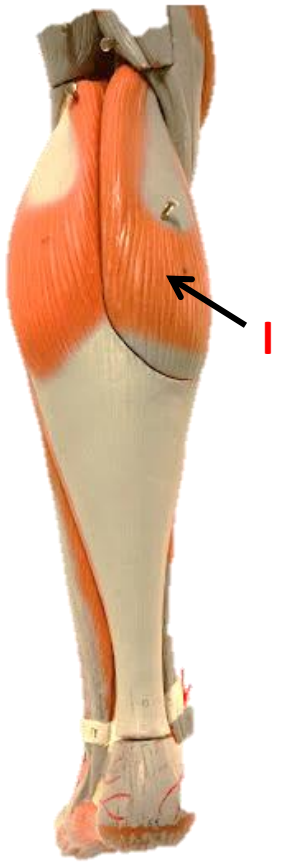
Posterior superficial



Posterior deep

Station 8: Musculature of the Lower Limb (continued)

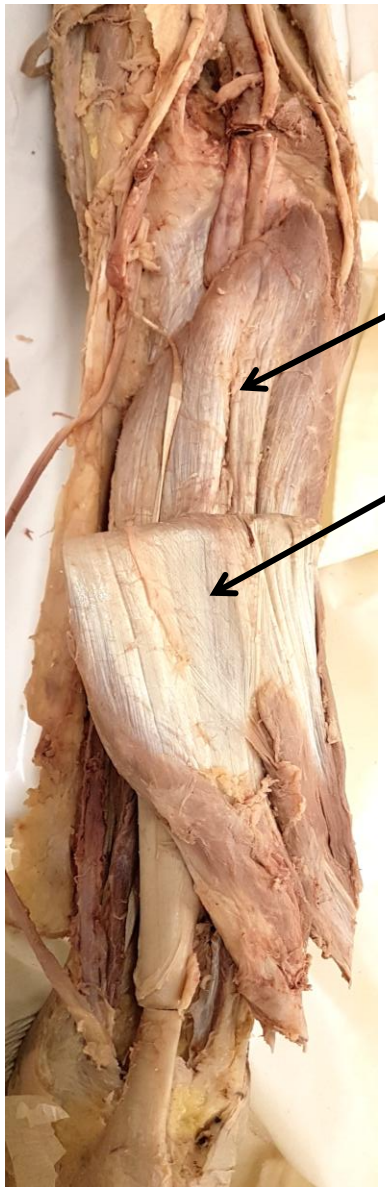
1. Identify and label the structures listed in the appendix on the diagrams below.



l



Posterior



m

l

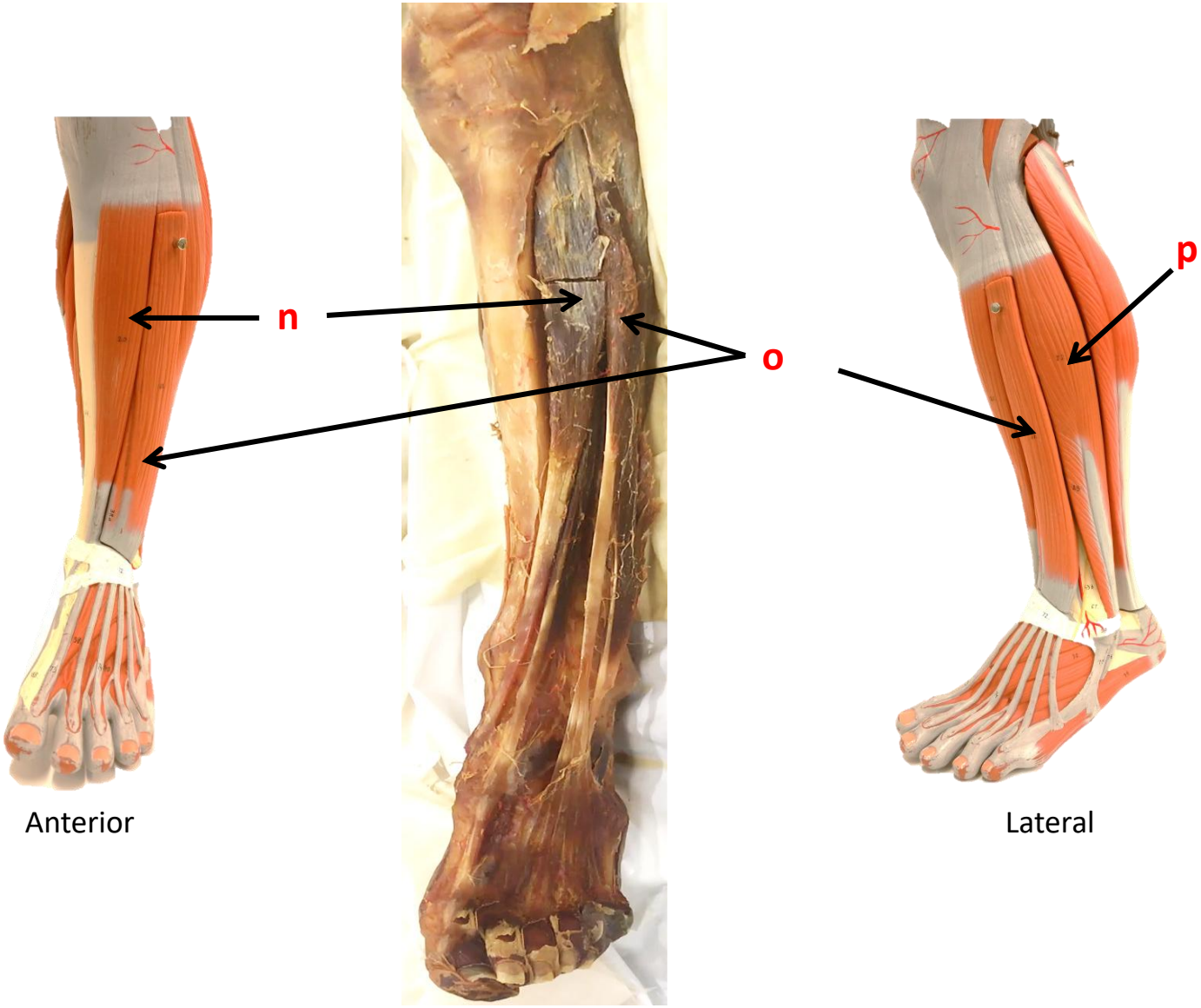


Lateral

Station 8: Musculature of the Lower Limb (continued)

1. Identify and label the structures listed in the appendix on the diagrams below.

Video Available:
Muscles of the anterior leg



Station 9: Practice Bellringer

In a Bellringer, you will be given a limited time to answer each question and identify the structure shown. Although you can go through this practice Bellringer at your own pace, it is recommended that you move through the questions as you would in a test situation.

Use the table below to record your answers.

1.		11.	
2.		12.	
3.		13.	
4.		14.	
5.		15.	
6.		16.	
7.		17.	
8.		18.	
9.		19.	
10.		20.	

Appendix

This list includes the terms you are responsible for knowing. Use it to guide you through the lab. Any of these structures may appear on the bellringer, possibly on different models or specimens than those used in the lab.

If a structure has a left and a right, be sure to include that in your answer.

Station 1: Muscle Tissue

Smooth muscle
Skeletal muscle
Cardiac muscle

Station 2: Structure of Skeletal Muscle

Sarcoplasmic reticulum
Transverse tubule
Sarcolemma
Sarcomere
Actin filament
Myosin filament
Myofibril

Station 3: General Musculature

Sternocleidomastoid
Platysma
Deltoid
Pectoralis major
Biceps brachii
External oblique
Rectus abdominis
Tensor fascia latae
Rectus femoris
Vastus lateralis
Vastus medialis
Adductor longus
Tibialis anterior
Trapezius
Triceps brachii
Latissimus dorsi
Gluteus maximus
Biceps femoris
Semitendinosus
Semimembranosus
Gastrocnemius
Adductor magnus

Station 4: Head and Neck Musculature

Platysma
Masseter
Sternocleidomastoid
Scalenes

Station 5: Anterior Thoracic and Abdominal Musculature

Pectoralis major
Pectoralis minor
Serratus anterior
Rectus abdominis
External oblique

Stations 6: Upper and Lower Back Musculature

Trapezius
Latissimus dorsi
Erector spinae
Iliopsoas
Quadratus lumborum

Station 7: Arm Musculature

Deltoid
Supraspinatus
Infraspinatus
Biceps brachii
Triceps brachii
Brachioradialis

Station 8: Lower Limb Musculature

Gluteus maximus
Piriformis
Rectus femoris
Vastus lateralis
Vastus medialis
Tensor fascia latae
Adductor longus
Adductor magnus
Biceps femoris
Semitendinosus
Semimembranosus
Tibialis anterior
Extensor digitorum longus
Fibularis longus
Gastrocnemius
Soleus

***Be aware when you pluralize a word it MAY change the spelling of the word. e.g., canaliculus becomes canaliculi**